KAIZEN MATHS

Lesson starter

Line and Curve Intersections

Mixed Conic Intersections

Name _____

S1

Find the intersection points:

Parabola: $y = 3x^2 + 2x - 1$

Line: $y = 2x - 4$

S2

Find the intersection points:

Circle: $(x - 1)^2 + (y - 0)^2 = 16$

Line: $y = 2x + 12$

S3

Find the intersection points:

Parabola: $y = 2x^2 - 2x - 2$

Line: $y = 3x + 2$

S4

Find the intersection points:

Circle: $(x - 2)^2 + (y - 1)^2 = 9$

Line: $y = 3x + 11$

Show your working

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ALG-5F81-L4 · S

Show your working

KAIZEN MATHS

Exit ticket

Line and Curve Intersections

Mixed Conic Intersections

Name _____

EA1

Find the intersection points:

Parabola: $y = 4x^2 - 2x - 3$

Line: $y = 2x - 3$

EA2

Find the intersection points:

Circle: $(x - 1)^2 + (y + 1)^2 = 16$

Line: $y = 0.577x + 3.041$

Show your working

KAIZEN MATHS

Exit ticket

Line and Curve Intersections

Mixed Conic Intersections

Name _____

EB1

Find the intersection points:

Parabola: $y = 2x^2 - 4x - 4$

Line: $y = -3x - 1$

Show your working

KAIZEN MATHS

Exit ticket

Line and Curve Intersections

Mixed Conic Intersections

Name _____

EA1

Find the intersection points:

Parabola: $y = 4x^2 - 2x - 3$

Line: $y = 2x - 3$

EA2

Find the intersection points:

Circle: $(x - 1)^2 + (y + 1)^2 = 16$

Line: $y = 0.577x + 3.041$

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Show your working

KAIZEN MATHS

Exit ticket

Line and Curve Intersections

Mixed Conic Intersections

Name _____

EB1

Find the intersection points:

Parabola: $y = 2x^2 - 4x - 4$

Line: $y = -3x - 1$

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Show your working

Teacher answers & working

Line and Curve Intersections · Mixed Conic Intersections

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S1

Find the intersection points:

Parabola: $y = 3x^2 + 2x - 1$

Line: $y = 2x - 4$

Answer:

No intersection points

Working:

Given equations

Parabola: $y = 3x^2 + 2x - 1$

Line: $y = 2x - 4$

Substitute the line into the conic

This gives the quadratic:

$3x^2 + 3 = 0$

Use the discriminant

$D = b^2 - 4ac = -36.000$

Since D < 0, there are no real intersection points.

S2

Find the intersection points:

Circle: $(x - 1)^2 + (y - 0)^2 = 16$

Line: $y = 2x + 12$

Answer:

No intersection points

Working:

Given equations

Circle: $(x - 1)^2 + (y - 0)^2 = 16$

Line: $y = 2x + 12$

Substitute the line into the conic

This gives the quadratic:

$5x^2 + 46x + 129 = 0$

Use the discriminant

$D = b^2 - 4ac = -464.000$

Since D < 0, there are no real intersection points.

Teacher answers & working

Line and Curve Intersections · Mixed Conic Intersections

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S3

Find the intersection points:

Parabola: $y = 2x^2 - 2x - 2$

Line: $y = 3x + 2$

Answer:

Two intersection points: (3.137, 11.412) and (-0.637, 0.088)

Working:

Given equations

Parabola: $y = 2x^2 - 2x - 2$

Line: $y = 3x + 2$

Substitute the line into the conic

This gives the quadratic:

$2x^2 - 5x - 4 = 0$

Use the discriminant

$D = b^2 - 4ac = 57.000$

Solve the quadratic to find x-values: 3.137, -0.637

Substitute each x-value into $y = 3x + 2$ to find y.

Point 1: x = 3.137, y = 11.412, so (3.137, 11.412)

Point 2: x = -0.637, y = 0.088, so (-0.637, 0.088)

S4

Find the intersection points:

Circle: $(x - 2)^2 + (y - 1)^2 = 9$

Line: $y = 3x + 11$

Answer:

No intersection points

Working:

Given equations

Circle: $(x - 2)^2 + (y - 1)^2 = 9$

Line: $y = 3x + 11$

Substitute the line into the conic

This gives the quadratic:

$10x^2 + 56x + 95 = 0$

Use the discriminant

$D = b^2 - 4ac = -664.000$

Since $D < 0$, there are no real intersection points.

Teacher answers & working

Line and Curve Intersections · Mixed Conic Intersections

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EA1

Find the intersection points:

Parabola: $y = 4x^2 - 2x - 3$

Line: $y = 2x - 3$

Answer:

Two intersection points: (1.000, -1.000) and (0.000, -3.000)

Working:

Given equations

Parabola: $y = 4x^2 - 2x - 3$

Line: $y = 2x - 3$

Substitute the line into the conic

This gives the quadratic:

$4x^2 - 4x = 0$

Use the discriminant

$D = b^2 - 4ac = 16.000$

Solve the quadratic to find x-values: 1.000, 0.000

Substitute each x-value into $y = 2x - 3$ to find y.

Point 1: x = 1.000, y = -1.000, so (1.000, -1.000)

Point 2: x = 0.000, y = -3.000, so (0.000, -3.000)

Teacher answers & working

Line and Curve Intersections · Mixed Conic Intersections

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EA2

Find the intersection points:

Circle: $(x - 1)^2 + (y + 1)^2 = 16$

Line: $y = 0.577x + 3.041$

Answer:

One intersection point: (-1.000, 2.464)

Working:

Given equations

Circle: $(x - 1)^2 + (y + 1)^2 = 16$

Line: $y = 0.577x + 3.041$

Substitute the line into the conic

This gives the quadratic:

$1.333x^2 + 2.667x + 1.333 = 0$

Use the discriminant

$D = b^2 - 4ac = 0.000$

Solve the quadratic to find x-values: -1.000

Substitute each x-value into $y = 0.577x + 3.041$ to find y.

Point 1: x = -1.000, y = 2.464, so (-1.000, 2.464)

Teacher answers & working

Line and Curve Intersections · Mixed Conic Intersections

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EB1

Find the intersection points:

Parabola: $y = 2x^2 - 4x - 4$

Line: $y = -3x - 1$

Answer:

Two intersection points: (1.500, -5.500) and (-1.000, 2.000)

Working:

Given equations

Parabola: $y = 2x^2 - 4x - 4$

Line: $y = -3x - 1$

Substitute the line into the conic

This gives the quadratic:

$2x^2 - x - 3 = 0$

Use the discriminant

$D = b^2 - 4ac = 25.000$

Solve the quadratic to find x-values: 1.500, -1.000

Substitute each x-value into $y = -3x - 1$ to find y.

Point 1: x = 1.500, y = -5.500, so (1.500, -5.500)

Point 2: x = -1.000, y = 2.000, so (-1.000, 2.000)